

Question	Answer	Marks
10(a)	speed is (directly) proportional to distance	M1
	speed is speed of recession of galaxy from an observer, and distance is the distance of the galaxy from the observer	A1
10(b)(i)	galaxy is receding from the Earth	B1
	observed wavelength is redshifted from emitted wavelength	B1
10(b)(ii)	$\Delta\lambda / \lambda = v / c$ $(4.91 - 4.62) / 4.62 = v / (3.00 \times 10^8)$	C1
	$v = 1.9 \times 10^7 \text{ m s}^{-1}$	A1
10(b)(iii)	wavelength (of maximum intensity) is inversely proportional to temperature	B1
	observed wavelength too high, so determined temperature too low	B1
10(c)	$v = H_0 d$	C1
	$d = (1.9 \times 10^7) / (2.3 \times 10^{-18})$	A1
	$= 8.3 \times 10^{24} \text{ m}$	